

Fraction Universe

Every graded set, every week — 6 weeks, 30 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

WEEK 6

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

One Whole, Cut Up

Answers are whole numbers — when a question asks for a numerator, type just that number.

Warm-Up 11 ITEMS • RECALL

1 Halves in one whole

2 Quarters in one whole

3 Eighths in one whole

4 $1/2 = ?/4 \rightarrow$ numerator

5 $1/2 = ?/8 \rightarrow$ numerator

6 $3/3 =$ how many wholes?

7 Thirds in one whole

8 Fifths in one whole

9 Sixths in one whole

10 $1/3 = ?/6 -$ numerator

11 $1/5 = ?/10 -$ numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. How many thirds in 2 wholes? (Three in each whole)

.....

13. $5/5 =$ how many wholes? (All five pieces)

.....

14. $7/4 -$ how many whole ones? (4/4 makes one whole)

.....

15. $7/4 -$ how many quarters left over? (7 - 4)

.....

16. $3/4 = ?/12 -$ numerator

.....

17. $2/5 = ?/20 -$ numerator

.....

18. $5/6 = ?/12 -$ numerator

.....

19. $7/8 = ?/16 -$ numerator

.....

20. $2/3 = ?/15 -$ numerator

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Same Number, New Name

Multiply top and bottom by the same thing and nothing about the amount changes.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2} = \frac{?}{6} \rightarrow$ numerator

2 $\frac{1}{3} = \frac{?}{6} \rightarrow$ numerator

3 $\frac{2}{3} = \frac{?}{6} \rightarrow$ numerator

4 $\frac{1}{4} = \frac{?}{8} \rightarrow$ numerator

5 $\frac{3}{4} = \frac{?}{8} \rightarrow$ numerator

6 $\frac{1}{5} = \frac{?}{10} \rightarrow$ numerator

7 $\frac{1}{2} = \frac{?}{10} \rightarrow$ numerator

8 $\frac{1}{4} = \frac{?}{12} \rightarrow$ numerator

9 $\frac{1}{3} = \frac{?}{9} \rightarrow$ numerator

10 $\frac{2}{3} = \frac{?}{9} \rightarrow$ numerator

11 $\frac{3}{5} = \frac{?}{10} \rightarrow$ numerator

12 $\frac{1}{6} = \frac{?}{12} \rightarrow$ numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $\frac{3}{4} = \frac{?}{12} \rightarrow$ numerator (The bottom was tripled) _____

14. $\frac{2}{5} = \frac{?}{20} \rightarrow$ numerator (The bottom was multiplied by 4) _____

15. $\frac{6}{8}$ in quarters = $\frac{?}{4} \rightarrow$ numerator (Divide both by 2) _____

16. $\frac{10}{15}$ simplified $\rightarrow$ numerator (Divide both by 5) _____

17. $\frac{9}{12}$ simplified $\rightarrow$ denominator (Divide both by 3) _____

18. $\frac{6}{8} = \frac{?}{4} \rightarrow$ numerator

19. $\frac{10}{15} = \frac{?}{3} \rightarrow$ numerator

20. $\frac{12}{16} = \frac{?}{4} \rightarrow$ numerator

21. $\frac{9}{15} = \frac{?}{5} \rightarrow$ numerator

22. $\frac{14}{21} = \frac{?}{3} \rightarrow$ numerator



Which Is Bigger

Make the pieces the same size first. Then it's just counting.

Warm-Up 12 ITEMS • RECALL

1 Bigger: $\frac{1}{2}$ or $\frac{1}{4}$ → its denominator

2 Bigger: $\frac{1}{3}$ or $\frac{1}{5}$ → its denominator

3 Bigger: $\frac{3}{4}$ or $\frac{1}{4}$ → its numerator

4 Bigger: $\frac{2}{5}$ or $\frac{3}{5}$ → its numerator

5 Bigger: $\frac{5}{6}$ or $\frac{5}{8}$ → its denominator

6 Bigger: $\frac{7}{10}$ or $\frac{3}{10}$ → its numerator

7 Bigger: $\frac{1}{2}$ or $\frac{1}{5}$ — its denominator

8 Bigger: $\frac{1}{4}$ or $\frac{1}{6}$ — its denominator

9 Bigger: $\frac{4}{5}$ or $\frac{2}{5}$ — its numerator

10 Bigger: $\frac{5}{8}$ or $\frac{3}{8}$ — its numerator

11 Bigger: $\frac{1}{3}$ or $\frac{1}{10}$ — its denominator

12 Bigger: $\frac{7}{10}$ or $\frac{3}{10}$ — its numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{2}{3}$ or $\frac{3}{5}$ in fifteenths → bigger numerator (10/15 against 9/15)

14. $\frac{3}{4}$ or $\frac{5}{8}$ in eighths → bigger numerator (6/8 against 5/8)

15. $\frac{5}{6}$ or $\frac{7}{9}$ in eighteenths → bigger numerator (15/18 against 14/18)

16. Closer to 1: $\frac{7}{8}$ or $\frac{5}{6}$ → its denominator (A gap of $\frac{1}{8}$ beats a gap of $\frac{1}{6}$)

17. $\frac{2}{3}$ and $\frac{3}{4}$ in twelfths → bigger numerator (8/12 against 9/12)

18. Bigger: $\frac{2}{3}$ or $\frac{3}{5}$ in fifteenths — its numerator

19. Bigger: $\frac{3}{4}$ or $\frac{5}{8}$ in eighths — its numerator

20. Bigger: $\frac{5}{6}$ or $\frac{4}{5}$ in thirtieths — its numerator

21. Bigger: $\frac{1}{2}$ or $\frac{4}{9}$ in eighteenths — its numerator

22. Bigger: $\frac{7}{12}$ or $\frac{1}{2}$ in twelfths — its numerator

26. Halfway between $\frac{1}{4}$ and $\frac{1}{2}$ as $\frac{?}{8}$ — numerator

Adding Needs Matching Pieces

Change one fraction, or both, until the bottoms agree. Then count.

Warm-Up 2 ITEMS • RECALL

1 $7/10 - 4/10 = ?/10 \rightarrow$
numerator

2 $7/10 - 3/10 = ?/10 -$
numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

3. A whole cut into 2 equal parts — one part is 1 over what

4. A whole cut into 3 equal parts — one part is 1 over what

5. A whole cut into 4 equal parts — one part is 1 over what

6. A whole cut into 6 equal parts — one part is 1 over what

7. A whole cut into 8 equal parts — one part is 1 over what

8. How many 2ths make one whole

9. How many 3ths make one whole

10. How many 4ths make one whole

11. How many 6ths make one whole

12. How many 8ths make one whole

13. Which is the bigger unit fraction: $1/2$ or $1/4$ — give the denominator

14. Which is the bigger unit fraction: $1/3$ or $1/6$ — give the denominator

15. Which is the bigger unit fraction: $1/4$ or $1/8$ — give the denominator

 Challenge NOT GRADED · REASON IT OUT

25. $\frac{1}{4} + \frac{1}{4} = \frac{?}{4} \rightarrow$ numerator

26. $\frac{2}{5} + \frac{1}{5} = \frac{?}{5} \rightarrow$ numerator

27. $\frac{3}{8} + \frac{2}{8} = \frac{?}{8} \rightarrow$ numerator

28. $\frac{5}{6} - \frac{2}{6} = \frac{?}{6} \rightarrow$ numerator

29. $\frac{3}{4} + \frac{1}{4} =$ how many wholes?

30. $\frac{1}{2} + \frac{1}{4} = \frac{?}{4} \rightarrow$ numerator $(\frac{2}{4} + \frac{1}{4})$

31. $\frac{2}{3} + \frac{1}{6} = \frac{?}{6} \rightarrow$ numerator ($\frac{4}{6} + \frac{1}{6}$)

32. $\frac{3}{4} - \frac{1}{8} = \frac{?}{8} \rightarrow$ numerator ($\frac{6}{8} - \frac{1}{8}$)

33. $\frac{5}{6} - \frac{1}{3} = \frac{?}{6} \rightarrow$ numerator $(\frac{5}{6} - \frac{2}{6})$

34. $\frac{1}{2} + \frac{1}{3} = \frac{?}{6} \rightarrow$ numerator $(\frac{3}{6} + \frac{2}{6})$

35. $\frac{3}{4} + \frac{2}{3} = \frac{?}{12} \rightarrow$ numerator ($\frac{9}{12} + \frac{8}{12}$)

36. $2 - 3/5 = ?/5 \rightarrow$ numerator $(10/5 - 3/5)$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Fraction of a Set & Closest to One

Enrichment day. Divide by the bottom, multiply by the top — always that order.

 Challenge NOT GRADED • REASON IT OUT

16. $\frac{1}{4}$ of 20 ($20 \div 4$)

17. $\frac{2}{5}$ of 30 ($30 \div 5 = 6$, then $\times 2$)

18. $\frac{3}{8}$ of 40 ($40 \div 8 = 5$, then $\times 3$)

19. $\frac{2}{3}$ of 45 ($45 \div 3 = 15$, then $\times 2$)

20. $\frac{5}{6}$ of 42 ($42 \div 6 = 7$, then $\times 5$)

21. $\frac{3}{4}$ of a number is 18. The number? ($18 \div 3 = 6$ is one quarter)

22. Closest to 1 without going over: $\frac{5}{6}, \frac{7}{8}, \frac{9}{10} \rightarrow$ its denominator (Gaps of $\frac{1}{6}, \frac{1}{8}, \frac{1}{10}$)

23. Eighths in 2 wholes and 3 eighths ($16 + 3$)

24. $\frac{1}{3}$ of 30

25. $\frac{1}{5}$ of 40

26. $\frac{2}{3}$ of 21

27. $\frac{3}{4}$ of 40

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • One Whole, Cut Up

OUT OF 20

Warm-Up • 1-11 • 2 | 4 | 8 | 2 | 4 | 1 | 3 | 5 | 6 | 2 | 2

Core • 12-20 • 6 | 1 | 1 | 3 | 9 | 8 | 10 | 14 | 10

Challenge • 21-26 • 3 | 4 | 2 | 10 | 7 | 5

1.2 • Same Number, New Name

OUT OF 22

Warm-Up • 1-12 • 3 | 2 | 4 | 2 | 6 | 2 | 5 | 3 | 3 | 6 | 6 | 2

Core • 13-22 • 9 | 8 | 3 | 2 | 4 | 3 | 2 | 3 | 3 | 2

Challenge • 23-26 • 2 | 9 | 6 | 5

1.3 • Which Is Bigger

OUT OF 22

Warm-Up • 1-12 • 2 | 3 | 3 | 3 | 6 | 7 | 2 | 4 | 4 | 5 | 3 | 7

Core • 13-22 • 10 | 6 | 15 | 8 | 9 | 10 | 6 | 25 | 9 | 7

Challenge • 23-26 • 21 | 5 | 16 | 3

1.4 • Adding Needs Matching Pieces

OUT OF 15

Warm-Up • 1-2 • 3 | 4

Core • 3-15 • 2 | 3 | 4 | 6 | 8 | 2 | 3 | 4 | 6 | 8 | 2 | 3 | 4

Challenge • 25-36 • 2 | 3 | 5 | 3 | 1 | 3 | 5 | 5 | 3 | 5 | 17 | 7

Fri • Fraction of a Set & Closest to One

ENRICHMENT

Challenge • 1-14 • 5 | 12 | 15 | 30 | 35 | 24 | 10 | 19 | 10 | 8 | 14 | 30 | 18
| 24

Same Number, New Clothes

Multiply top and bottom by the same thing and the value never moves. Type fractions like $\frac{3}{4}$.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2} = \frac{?}{4}$ — the numerator

2 $\frac{1}{2} = \frac{?}{6}$

3 $\frac{1}{3} = \frac{?}{6}$

4 $\frac{1}{4} = \frac{?}{8}$

5 $\frac{2}{3} = \frac{?}{6}$

6 $\frac{3}{4} = \frac{?}{8}$

7 $\frac{1}{2} = \frac{?}{8}$

8 $\frac{1}{3} = \frac{?}{12}$

9 $\frac{1}{5} = \frac{?}{15}$

10 $\frac{2}{3} = \frac{?}{12}$

11 $\frac{3}{5} = \frac{?}{15}$

12 $\frac{1}{4} = \frac{?}{16}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{2}{5} = \frac{?}{15}$

14. $\frac{3}{8} = \frac{?}{24}$

15. $\frac{5}{6} = \frac{?}{18}$

16. $\frac{4}{7} = \frac{?}{21}$

17. $\frac{7}{10} = \frac{?}{100}$

18. $\frac{4}{5} = \frac{?}{25}$

19. $\frac{5}{8} = \frac{?}{24}$

20. $\frac{3}{7} = \frac{?}{21}$

21. $\frac{5}{9} = \frac{?}{27}$



Working space



Error journal

Simplest Form

Divide top and bottom by the same thing until nothing else fits.

Warm-Up 11 ITEMS • RECALL

1 $\frac{2}{4}$ in simplest form —
type as a/b

2 $\frac{3}{6}$

3 $\frac{4}{8}$

4 $\frac{2}{6}$

5 $\frac{5}{10}$

6 $\frac{3}{9}$

7 $\frac{6}{12}$ in simplest form —
type as a/b

8 $\frac{4}{6}$

9 $\frac{5}{15}$

10 $\frac{8}{10}$

11 $\frac{9}{12}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. $\frac{6}{8}$

13. $\frac{10}{15}$

14. $\frac{12}{16}$

15. $\frac{18}{24}$

16. $\frac{20}{25}$

17. $\frac{21}{28}$

18. $\frac{30}{45}$

19. $\frac{14}{35}$

20. $\frac{27}{36}$

Spot the Equivalent

Cross-multiply, or find a common denominator. Either proof counts.

Warm-Up 12 ITEMS • RECALL

1 Is $1/2$ the same as $2/4$ — yes or no

2 Is $1/3$ the same as $2/6$

3 Is $1/2$ the same as $1/3$

4 Is $3/4$ the same as $6/8$

5 Is $2/5$ the same as $4/10$

6 Is $2/3$ the same as $3/4$

7 Is $1/4$ the same as $2/8$ — yes or no

8 Is $2/3$ the same as $4/6$

9 Is $1/2$ the same as $2/5$

10 Is $3/5$ the same as $6/10$

11 Is $1/3$ the same as $3/9$

12 Is $2/4$ the same as $3/5$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is $6/9$ the same as $8/12$ — yes or no

.....

14. Is $5/8$ the same as $10/18$

.....

15. Is $9/12$ the same as $15/20$

.....

16. $1/2 \times 5/5$ — type as a/b

.....

17. Is $7/21$ the same as $1/3$

.....

18. Is $6/9$ the same as $8/12$

.....

19. Is $5/8$ the same as $10/16$

.....

20. Is $4/7$ the same as $8/15$

.....

21. Is $9/15$ the same as $12/20$

.....

22. Is $7/9$ the same as $14/17$

.....

Fractions Equal to One

Any fraction with matching top and bottom is just 1 wearing a costume.

Warm-Up 12 ITEMS • RECALL

1 $5/5$ as a whole number2 $8/8$ 3 $12/12$ 4 $4/4$ 5 $100/100$ 6 $2/2$ 7 $7/7$ as a whole number8 $9/9$ 9 $6/6$ 10 $20/20$ 11 $3/3$ 12 $50/50$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $6/3$ as a whole number14. $12/4$ 15. $20/5$

16. How many quarters make 2

17. How many thirds make 3

18. $10/5$ as a whole number19. $18/6$ 20. $24/8$ 21. $9/2$ as a mixed number — the whole part22. $13/5$ — the whole part



Working space



Error journal

Equivalence Wall

Build the strips and find every pair that lines up.

Warm-Up 3 ITEMS • RECALL

1 Halves in a whole

2 Quarters in a whole

3 Sixths in a whole

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

4. Quarters that line up with $\frac{1}{2}$ — type as a/b

.....

5. Sixths that line up with $\frac{1}{3}$

.....

6. $\frac{1}{2} = ?/12$

.....

7. $\frac{2}{3} = ?/12$

.....

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Same Number, New Clothes

OUT OF 21

Warm-Up • 1-12 • 2 | 3 | 2 | 2 | 4 | 6 | 4 | 4 | 3 | 8 | 9 | 4

Core • 13-21 • 6 | 9 | 15 | 12 | 70 | 20 | 15 | 9 | 15

Challenge • 22-25 • $\frac{3}{4}$ | 7 | $\frac{3}{4}$ | 8

2.2 • Simplest Form

OUT OF 20

Warm-Up • 1-11 • $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{4}{5}$ | $\frac{3}{4}$

Core • 12-20 • $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{4}{5}$ | $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{2}{5}$ | $\frac{3}{4}$

Challenge • 21-24 • $\frac{3}{4}$ | 12 | $\frac{3}{4}$ | 15

2.3 • Spot the Equivalent

OUT OF 22

Warm-Up • 1-12 • yes | yes | no | yes | yes | no | yes | yes | no | yes | yes | no

Core • 13-22 • yes | no | yes | $\frac{5}{10}$ | yes | yes | yes | no | yes | no

Challenge • 23-26 • 9 | yes | 6 | yes

2.4 • Fractions Equal to One

OUT OF 22

Warm-Up • 1-12 • 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1

Core • 13-22 • 2 | 3 | 4 | 8 | 9 | 2 | 3 | 3 | 4 | 2

Challenge • 23-26 • 3 | 2 | 5 | 3

Fri • Equivalence Wall

OUT OF 8

Warm-Up • 1-4 • 2 | 4 | 6 | 10

Core • 5-8 • $\frac{2}{4}$ | $\frac{2}{6}$ | 6 | 8

Challenge • 9-14 • $\frac{9}{12}$ | $\frac{8}{12}$ | 12 | 12 | $\frac{2}{3}$ | 7

Common Denominators

Cut both into the same size pieces, then just compare the counts.

Warm-Up 11 ITEMS • RECALL

1 Common denominator of $\frac{1}{2}$ and $\frac{1}{4}$

2 Common denominator of $\frac{1}{3}$ and $\frac{1}{6}$

3 Common denominator of $\frac{1}{2}$ and $\frac{1}{3}$

4 $\frac{1}{2}$ as quarters — the numerator

5 $\frac{1}{3}$ as sixths — the numerator

6 Common denominator of $\frac{1}{4}$ and $\frac{1}{8}$

7 Common denominator of $\frac{1}{2}$ and $\frac{1}{6}$

8 Common denominator of $\frac{1}{3}$ and $\frac{1}{9}$

9 $\frac{1}{2}$ as sixths — the numerator

10 $\frac{1}{3}$ as ninths — the numerator

11 $\frac{1}{4}$ as eighths — the numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Common denominator of $\frac{2}{3}$ and $\frac{3}{4}$

13. Common denominator of $\frac{3}{5}$ and $\frac{1}{2}$

14. $\frac{2}{3}$ as twelfths — the numerator

15. $\frac{3}{4}$ as twelfths — the numerator

16. Common denominator of $\frac{5}{6}$ and $\frac{3}{8}$

17. Smallest common denominator of $\frac{1}{4}$ and $\frac{1}{6}$

18. Of $\frac{1}{3}$ and $\frac{1}{5}$

19. Of $\frac{2}{5}$ and $\frac{3}{10}$

20. Of $\frac{3}{8}$ and $\frac{1}{6}$

21. Of $\frac{5}{12}$ and $\frac{1}{8}$

★ Challenge NOT GRADED • REASON IT OUT

22. Smallest common denominator of $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{1}{8}$

23. Common denominator of $\frac{7}{10}$ and $\frac{2}{15}$

24. Smallest common denominator of $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{6}$

25. Common denominator of $\frac{5}{9}$ and $\frac{7}{12}$

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Benchmarks

Nearer 0, a half, or 1? Decide before you calculate anything.

Warm-Up 12 ITEMS • RECALL

1 Is $\frac{1}{8}$ nearer 0 or 1 — type 0 or 1

2 Is $\frac{7}{8}$ nearer 0 or 1

3 $\frac{4}{8}$ simplified — type as a/b

4 Is $\frac{5}{6}$ nearer $\frac{1}{2}$ or 1 — type $\frac{1}{2}$ or 1

5 Is $\frac{1}{5}$ nearer 0 or $\frac{1}{2}$ — type 0 or $\frac{1}{2}$

6 Is $\frac{3}{5}$ more or less than $\frac{1}{2}$ — type more or less

7 Is $\frac{1}{10}$ nearer 0 or 1 — type 0 or 1

8 Is $\frac{9}{10}$ nearer 0 or 1

9 $\frac{5}{10}$ simplified — type as a/b

10 Is $\frac{1}{6}$ nearer 0 or $\frac{1}{2}$ — type 0 or $\frac{1}{2}$

11 Is $\frac{3}{5}$ nearer $\frac{1}{2}$ or 1 — type $\frac{1}{2}$ or 1

12 $\frac{6}{12}$ simplified — type as a/b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is $\frac{5}{9}$ more or less than $\frac{1}{2}$

.....

14. Is $\frac{4}{9}$ more or less than $\frac{1}{2}$

.....

15. Is $\frac{7}{16}$ more or less than $\frac{1}{2}$

.....

16. Is $\frac{9}{16}$ more or less than $\frac{1}{2}$

.....

17. Nearest benchmark to $\frac{11}{12}$ — type 0, $\frac{1}{2}$ or 1

.....

18. Is $\frac{11}{20}$ more or less than $\frac{1}{2}$

.....

19. Nearest benchmark to $\frac{9}{10}$ — type 0, $\frac{1}{2}$ or 1

.....

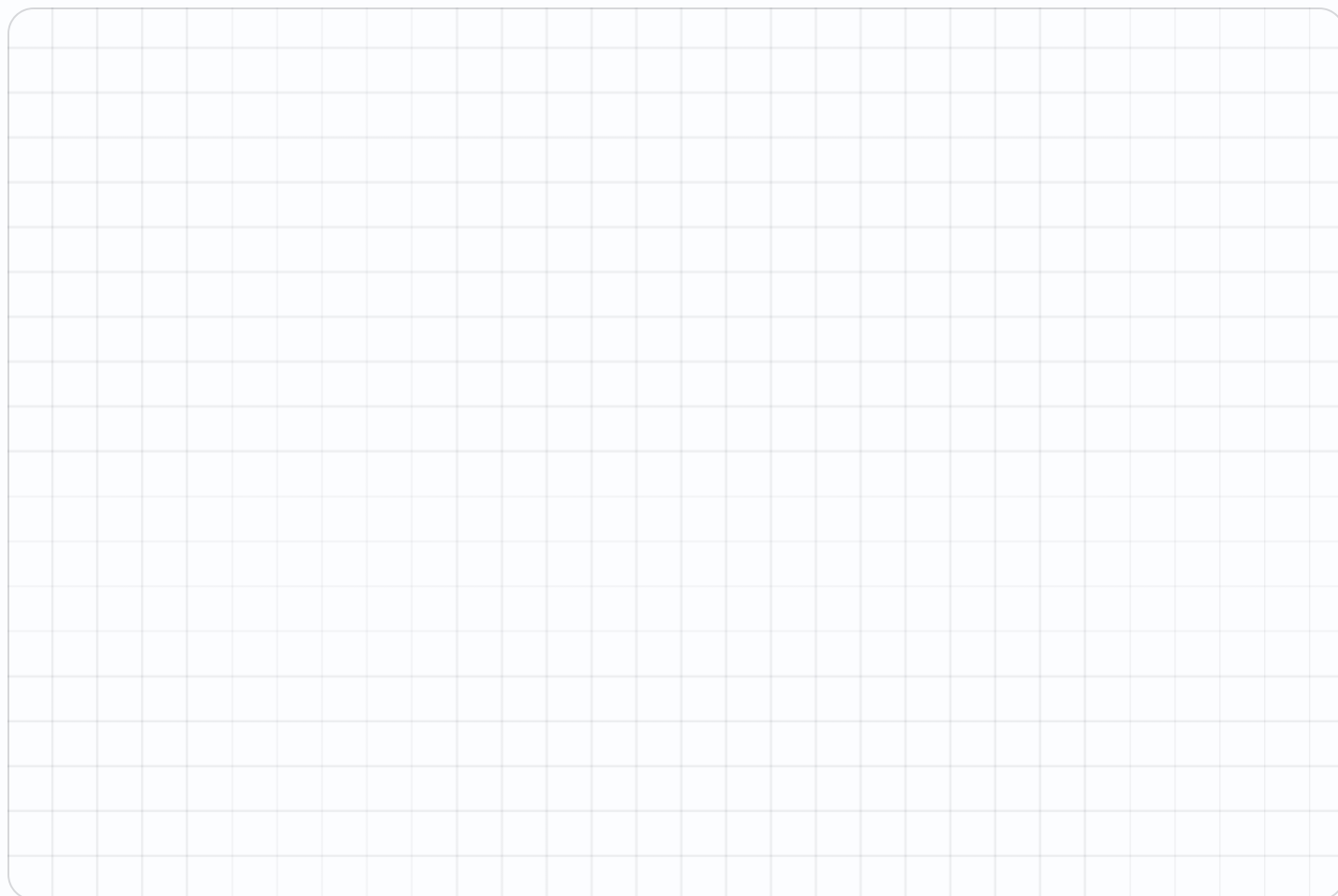
★ Challenge NOT GRADED • REASON IT OUT

20. Is $13/25$ more or less than $1/2$

21. Nearest benchmark to $8/15$ — type 0, $1/2$ or 1

22. Is $17/35$ more or less than $1/2$

23. Nearest benchmark to $5/11$ — type 0, $1/2$ or 1

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Comparing & Ordering

Common denominator, or reason from benchmarks. Say which you used.

Warm-Up 12 ITEMS • RECALL

1 Larger: $\frac{1}{2}$ or $\frac{1}{3}$ — type it

2 Larger: $\frac{3}{4}$ or $\frac{1}{2}$

3 Larger: $\frac{1}{4}$ or $\frac{1}{8}$

4 Larger: $\frac{2}{3}$ or $\frac{1}{3}$

5 Smaller: $\frac{1}{5}$ or $\frac{1}{6}$

6 Larger: $\frac{5}{8}$ or $\frac{3}{8}$

7 Larger: $\frac{1}{2}$ or $\frac{1}{4}$ — type it

8 Larger: $\frac{2}{3}$ or $\frac{1}{2}$

9 Larger: $\frac{3}{5}$ or $\frac{2}{5}$

10 Larger: $\frac{1}{3}$ or $\frac{1}{4}$

11 Smaller: $\frac{1}{8}$ or $\frac{1}{10}$

12 Larger: $\frac{5}{6}$ or $\frac{1}{2}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Larger: $\frac{2}{3}$ or $\frac{3}{4}$ — type it

14. Larger: $\frac{3}{5}$ or $\frac{5}{8}$

15. Larger: $\frac{5}{6}$ or $\frac{7}{9}$

16. Smallest of $\frac{1}{3}$, $\frac{2}{5}$, $\frac{1}{4}$ — type it

17. Largest of $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{12}$

18. Larger: $\frac{2}{3}$ or $\frac{5}{8}$ — type it

19. Larger: $\frac{3}{7}$ or $\frac{1}{2}$

20. Larger: $\frac{7}{12}$ or $\frac{5}{9}$

21. Larger: $\frac{4}{5}$ or $\frac{7}{9}$

22. Smaller: $\frac{5}{12}$ or $\frac{3}{8}$

 Challenge NOT GRADED • REASON IT OUT

23. Order $\frac{3}{4}$, $\frac{5}{6}$, $\frac{7}{8}$ — type the largest

24. Larger: $11/15$ or $7/10$ — type it

25. Order $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{10}$ — type the largest

26. Larger: $\frac{9}{14}$ or $\frac{2}{3}$ — type it

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

On the Number Line

Every fraction has one address between 0 and 1. Find it.

Warm-Up 9 ITEMS • RECALL

1 Halfway between 0 and 1
— type as a/b

2 Halfway between 0 and $\frac{1}{2}$

3 Halfway between $\frac{1}{2}$ and 1

4 Marks between 0 and 1 if
split into quarters

5 Marks if split into fifths

6 Halfway between 0 and $\frac{1}{4}$ — type as a/b

7 Halfway between $\frac{1}{4}$ and $\frac{1}{2}$

8 Halfway between $\frac{3}{4}$ and 1

9 Marks between 0 and 1 if
split into fifths

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. Halfway between $\frac{1}{4}$ and $\frac{1}{2}$ — type as a/b

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 $1/2 = ?/8$ — the numerator

2 $4/8$ simplified — type as a/b

3 $1/3 = ?/9$ — the numerator

4 $6/8$ simplified — type as a/b

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $6/8$ simplified

6. Common denominator of $2/3$ and $3/4$

7. Larger: $3/5$ or $5/8$ — type it

8. Is $5/9$ more or less than $1/2$

9. Larger: $2/3$ or $5/8$ — type it

10. Smallest common denominator of $1/4$ and $1/6$

★ Challenge NOT GRADED • REASON IT OUT

11. $\frac{3}{4}$ as a decimal

12. Order $\frac{3}{4}$, $\frac{5}{6}$, $\frac{7}{8}$ — type the largest

13. $\frac{3}{8}$ as a decimal

14. Is $\frac{17}{35}$ more or less than $\frac{1}{2}$

15. $48/64$ in simplest form — type as a/b



Working space



Error journal

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Common Denominators

OUT OF 21

Warm-Up • 1-11 • 4 | 6 | 6 | 2 | 2 | 8 | 6 | 9 | 3 | 3 | 2

Core • 12-21 • 12 | 10 | 8 | 9 | 24 | 12 | 15 | 10 | 24 | 24

Challenge • 22-25 • 24 | 30 | 12 | 36

3.2 • Benchmarks

OUT OF 19

Warm-Up • 1-12 • 0 | 1 | $\frac{1}{2}$ | 1 | 0 | more | 0 | 1 | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | $\frac{1}{2}$

Core • 13-19 • more | less | less | more | 1 | more | 1

Challenge • 20-23 • more | $\frac{1}{2}$ | less | $\frac{1}{2}$

3.3 • Comparing & Ordering

OUT OF 22

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{1}{4}$ | $\frac{2}{3}$ | $\frac{1}{6}$ | $\frac{5}{8}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{3}{5}$ | $\frac{1}{3}$ | $\frac{1}{10}$ | $\frac{5}{6}$

Core • 13-22 • $\frac{3}{4}$ | $\frac{5}{8}$ | $\frac{5}{6}$ | $\frac{1}{4}$ | $\frac{2}{3}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{7}{12}$ | $\frac{4}{5}$ | $\frac{3}{8}$

Challenge • 23-26 • $\frac{7}{8}$ | $\frac{11}{15}$ | $\frac{7}{10}$ | $\frac{2}{3}$

3.4 • On the Number Line

OUT OF 10

Warm-Up • 1-9 • $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | 3 | 4 | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{7}{8}$ | 4

Core • 10 • $\frac{3}{8}$

Challenge • 11-21 • 0.5 | 0.75 | 0.25 | 0.2 | 0.6 | 0.875 | $\frac{1}{2}$ | 0.125 | 0.1 | 0.375 | $\frac{1}{2}$

Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 4 | $\frac{1}{2}$ | 3 | $\frac{3}{4}$

Core • 5-11 • $\frac{3}{4}$ | 12 | $\frac{5}{8}$ | more | 0.75 | $\frac{2}{3}$ | 12

Challenge • 12-15 • $\frac{7}{8}$ | 0.375 | less | $\frac{3}{4}$

Adding Like Denominators

Same size pieces already. Add the counts, leave the bottom alone.

Warm-Up 3 ITEMS • RECALL

1 $\frac{3}{10} + \frac{2}{10}$

2 $\frac{1}{10} + \frac{4}{10}$

3 $\frac{1}{12} + \frac{5}{12}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

4. $\frac{5}{12} + \frac{3}{12}$

5. $\frac{7}{10} + \frac{1}{10}$

6. $\frac{4}{15} + \frac{6}{15}$

7. $\frac{7}{20} + \frac{3}{20}$

8. $\frac{3}{14} + \frac{4}{14}$

20. $\frac{2}{7} + \frac{3}{7}$

Subtracting Like Denominators

Same move, other direction.

✦ Warm-Up 2 ITEMS • RECALL

1 $9/10 - 4/10$

2 $8/10 - 3/10$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

3. $1 - 1/4$

4. $1 - 3/8$

5. $11/12 - 5/12$

6. $1 - 2/5$

7. $7/12 - 1/12$

8. $11/15 - 1/15$

9. $13/20 - 3/20$

Unlike Denominators

The grade-five ceiling. Convert first, then add.

Warm-Up 5 ITEMS • RECALL

1 A whole cut into 4 equal parts — one part is 1 over what

2 How many quarters make one whole

3 How many sixths make one whole

4 $\frac{2}{4}$ in its simplest form — the numerator

5 Is $\frac{3}{4}$ more or less than a half

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

6. $\frac{1}{2} = \frac{?}{4}$ — give the numerator

.....

7. $\frac{1}{3} = \frac{?}{6}$ — give the numerator

.....

8. $\frac{1}{4} = \frac{?}{8}$ — give the numerator

.....

9. $\frac{2}{3} = \frac{?}{6}$ — give the numerator

.....

10. $\frac{3}{4} = \frac{?}{8}$ — give the numerator

.....

11. $\frac{2}{4}$ in its simplest form — give the numerator

.....

12. $\frac{3}{6}$ in its simplest form — give the numerator

.....

13. $\frac{4}{8}$ in its simplest form — give the numerator

.....

14. $\frac{2}{6}$ in its simplest form — give the numerator

.....

15. How many 2ths equal one half

.....

16. How many 4ths equal one half

.....

17. How many 8ths equal one half

.....

Fractions in Words

Read it twice, then decide whether it is an add or a subtract.

Warm-Up 11 ITEMS • RECALL

1 Ate $\frac{1}{4}$ of a pizza, then $\frac{1}{4}$ more — total, as a/b

2 Had $\frac{3}{4}$, ate $\frac{1}{4}$ — left

3 $\frac{1}{2}$ hour plus $\frac{1}{2}$ hour — hours

4 $\frac{2}{5}$ of a jug plus $\frac{1}{5}$ — as a/b

5 1 whole minus $\frac{1}{3}$ — as a/b

6 Drank $\frac{1}{3}$ of a bottle, then $\frac{1}{3}$ more — total, as a/b

7 Had $\frac{5}{6}$, used $\frac{1}{6}$ — left

8 $\frac{1}{4}$ hour plus $\frac{3}{4}$ hour — hours

9 $\frac{3}{8}$ of a tank plus $\frac{1}{8}$ — as a/b

10 Read $\frac{1}{2}$ a book, then $\frac{1}{4}$ — as a/b

11 Had $\frac{7}{8}$, gave away $\frac{3}{8}$ — as a/b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. Walked $\frac{1}{2}$ mile then $\frac{1}{4}$ mile — total, as a/b

13. A $\frac{3}{4}$ cup recipe, you have $\frac{1}{2}$ — how much more, as a/b

14. Read $\frac{2}{3}$ of a book, then $\frac{1}{6}$ more — total, as a/b

15. Ran $\frac{5}{6}$ km and walked $\frac{1}{3}$ km — total, as a/b

16. 1 hour minus $\frac{1}{4}$ hour — minutes

17. Walked $\frac{2}{5}$ km then $\frac{1}{2}$ km — total as a/b

18. $\frac{3}{4}$ of an hour in minutes

19. $\frac{1}{3}$ of an hour in minutes

20. Ate $\frac{3}{8}$ of a pie, dad ate $\frac{1}{4}$ — total as a/b

21. $1 - \frac{5}{8}$ as a/b



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Closest to One

Draw cards, build a fraction, get nearest to 1 without going over.

✦ Warm-Up 4 ITEMS • RECALL

1 Which is nearer 1: $\frac{3}{4}$ or $\frac{1}{2}$ — type it

2 Which is nearer 1: $\frac{7}{8}$ or $\frac{3}{4}$

3 Which is nearer 1: $\frac{5}{6}$ or $\frac{2}{3}$ — type it

4 Which is nearer 1: $\frac{9}{10}$ or $\frac{4}{5}$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. From digits 5 and 6, the fraction under 1 — type as a/b

6. From 7 and 8, the fraction under 1

7. Which is nearer 1: $\frac{7}{8}$ or $\frac{8}{9}$ — type it

8. Which is nearer 1: $\frac{11}{12}$ or $\frac{5}{6}$



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Adding Like Denominators

OUT OF 8

Warm-Up • 1-3 • $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$

Core • 4-8 • $\frac{2}{3}$ | $\frac{4}{5}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$

Challenge • 9-20 • $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{2}{3}$ | 1 | $\frac{3}{2}$ | $\frac{7}{5}$ | $\frac{1}{3}$ | $\frac{5}{7}$

4.2 • Subtracting Like Denominators

OUT OF 9

Warm-Up • 1-2 • $\frac{1}{2}$ | $\frac{1}{2}$

Core • 3-9 • $\frac{3}{4}$ | $\frac{5}{8}$ | $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$

Challenge • 10-21 • $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{2}{3}$ | $\frac{5}{4}$ | $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{3}{7}$ | $\frac{1}{2}$ | $\frac{1}{3}$

4.3 • Unlike Denominators

OUT OF 17

Warm-Up • 1-5 • 4 | 4 | 6 | 1 | more

Core • 6-17 • 2 | 2 | 2 | 4 | 6 | 1 | 1 | 1 | 1 | 1 | 2 | 4

Challenge • 27-38 • $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{5}{8}$ | $\frac{3}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{11}{12}$ | $\frac{5}{12}$ | $\frac{1}{3}$ | $\frac{17}{12}$

4.4 • Fractions in Words

OUT OF 21

Warm-Up • 1-11 • $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | $\frac{3}{5}$ | $\frac{2}{3}$ | $\frac{2}{3}$ | $\frac{2}{3}$ | 1 | $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{1}{2}$

Core • 12-21 • $\frac{3}{4}$ | $\frac{1}{4}$ | $\frac{5}{6}$ | $\frac{7}{6}$ | 45 | $\frac{9}{10}$ | 45 | 20 | $\frac{5}{8}$ | $\frac{3}{8}$

Challenge • 22-26 • $\frac{1}{4}$ | $\frac{9}{8}$ | 40 | $\frac{8}{5}$ | 50

Fri • Closest to One

OUT OF 8

Warm-Up • 1-4 • $\frac{3}{4}$ | $\frac{7}{8}$ | $\frac{5}{6}$ | $\frac{9}{10}$

Core • 5-8 • $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{8}{9}$ | $\frac{11}{12}$

Challenge • 9-14 • $\frac{9}{10}$ | $\frac{1}{10}$ | $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{4}{5}$ | $\frac{1}{9}$

Fraction of a Set

Divide by the bottom, multiply by the top.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2}$ of 10

2 $\frac{1}{3}$ of 12

3 $\frac{1}{4}$ of 20

4 $\frac{1}{5}$ of 25

5 $\frac{1}{2}$ of 30

6 $\frac{1}{6}$ of 30

7 $\frac{1}{2}$ of 14

8 $\frac{1}{3}$ of 15

9 $\frac{1}{4}$ of 24

10 $\frac{1}{5}$ of 35

11 $\frac{1}{6}$ of 36

12 $\frac{1}{8}$ of 40

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

13. $\frac{2}{5}$ of 30

.....

14. $\frac{3}{4}$ of 20

.....

15. $\frac{2}{3}$ of 18

.....

16. $\frac{5}{6}$ of 24

.....

17. $\frac{3}{8}$ of 40

.....

18. $\frac{3}{4}$ of 32

.....



Of Money and Time

The same operator, applied to dollars and minutes.

✦ Warm-Up 9 ITEMS • RECALL

1 $\frac{1}{2}$ of \$20

2 $\frac{1}{4}$ of \$40

3 $\frac{1}{2}$ of an hour in minutes

4 $\frac{1}{4}$ of an hour in minutes

5 $\frac{1}{3}$ of \$30

6 $\frac{1}{5}$ of \$50

7 $\frac{1}{2}$ of \$10

8 $\frac{1}{4}$ of \$20

9 $\frac{1}{5}$ of \$25

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. $\frac{3}{4}$ of \$60

.....

11. $\frac{2}{3}$ of an hour in minutes

.....

12. $\frac{3}{5}$ of \$45

.....

13. $\frac{5}{6}$ of an hour in minutes

.....

14. $\frac{2}{5}$ of \$75

.....

15. $\frac{2}{3}$ of \$45

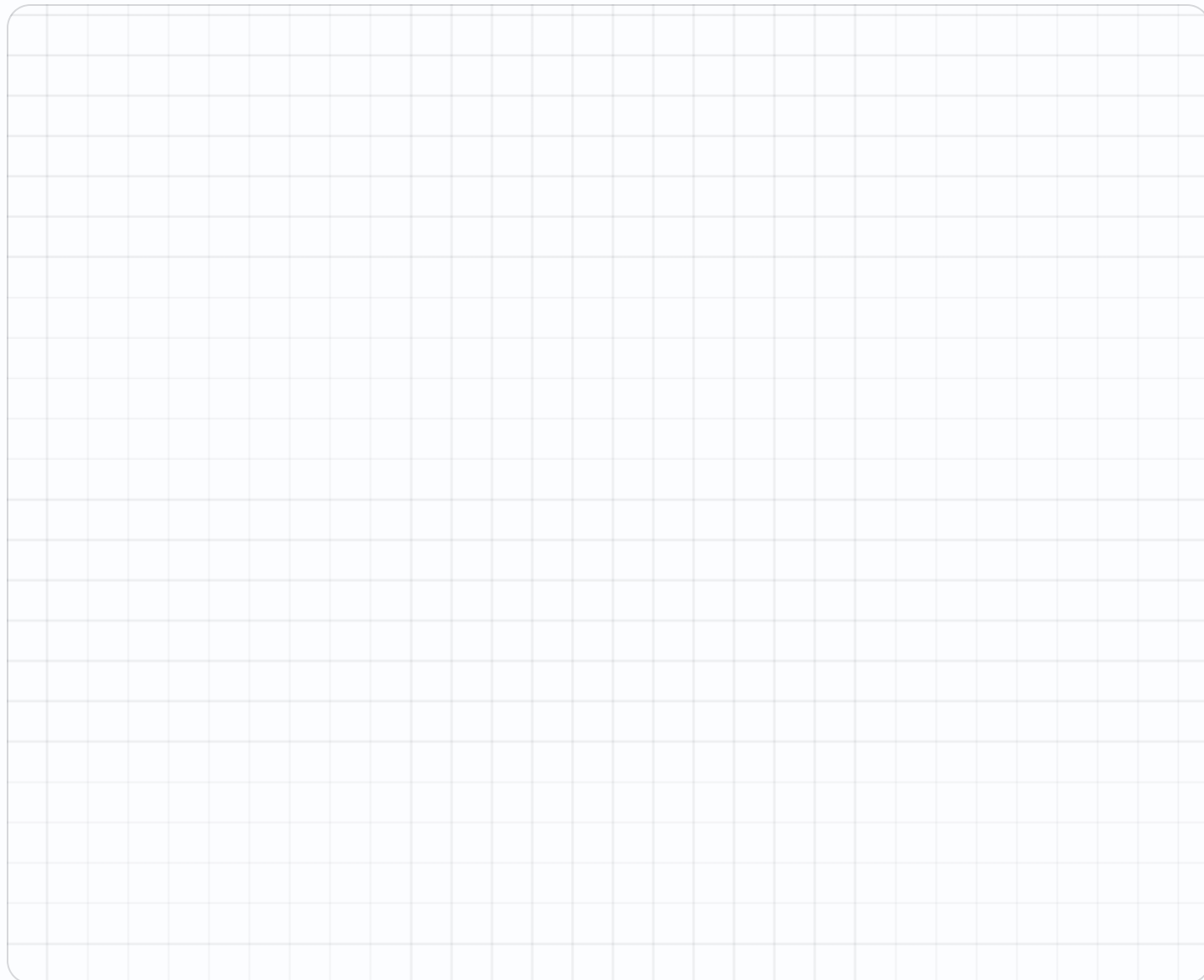
.....

16. $\frac{3}{4}$ of an hour in minutes

.....

17. $\frac{2}{5}$ of \$50

.....

★ Challenge NOT GRADED • REASON IT OUT18. $\frac{3}{8}$ of \$9619. $\frac{7}{12}$ of an hour in minutes**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Working Backwards

Given the part, find the whole.

Warm-Up 12 ITEMS • RECALL

1 Half a number is 6. The number

2 A third is 4. The number

3 A quarter is 5. The number

4 A fifth is 3. The number

5 Half is 15. The number

6 A sixth is 2. The number

7 $\frac{1}{2}$ of 100

8 $\frac{1}{4}$ of 100

9 $\frac{1}{10}$ of 100

10 $\frac{1}{5}$ of 100

11 $\frac{3}{4}$ of 100

12 $\frac{1}{2}$ of 60

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{2}{3}$ of a number is 12. The number
.....

14. $\frac{3}{4}$ of a number is 15. The number
.....

15. $\frac{2}{5}$ of a number is 10. The number
.....

16. $\frac{5}{6}$ of a number is 20. The number
.....

17. $\frac{3}{8}$ of a number is 9. The number
.....

18. $\frac{2}{5}$ of 200
.....

19. $\frac{3}{10}$ of 150
.....

20. $\frac{5}{8}$ of 160
.....

21. $\frac{2}{3}$ of 150
.....

22. $\frac{4}{5}$ of 250
.....



Working space



Error journal

Mixed Fraction Problems

No signposting. Read, decide, then compute.

✦ Warm-Up 10 ITEMS • RECALL

1 $\frac{1}{2}$ of 16
_____2 $1 - \frac{1}{2}$ — as a/b
_____3 $\frac{2}{6}$ simplified — as a/b
_____4 $\frac{1}{3}$ of 15
_____5 $\frac{1}{2}$ of 8
_____6 $\frac{1}{3}$ of 9
_____7 $\frac{1}{4}$ of 16
_____8 $\frac{1}{2}$ of 22
_____9 $\frac{1}{5}$ of 20
_____10 $\frac{1}{6}$ of 24

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $\frac{3}{4}$ of 24
.....12. $\frac{2}{5}$ of a number is 8. The number
.....13. Larger: $\frac{5}{8}$ or $\frac{2}{3}$ — type it
.....14. $\frac{3}{4}$ of 28
.....15. $\frac{2}{3}$ of 33
.....16. $\frac{4}{5}$ of 45
.....17. $\frac{5}{8}$ of 48
.....18. $\frac{3}{7}$ of 42
.....

Fold the Universe, Built

Finish the paper number line with every fraction landed on it.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

Warm-Up 4 ITEMS • RECALL

- 1 Folds to get halves

- 2 Folds to get quarters

- 3 $\frac{1}{2}$ of 12

- 4 $\frac{1}{4}$ of 12

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Folds to get eighths
.....
6. Sections after 4 folds
.....
7. $\frac{2}{3}$ of 24
.....
8. $\frac{3}{5}$ of 40
.....

 Challenge NOT GRADED • REASON IT OUT

9. Sections after 6 folds

10. Folds needed for 32 sections

11. Marks strictly between 0 and 1 on a sixteenths line

12. $\frac{7}{8}$ of 96

13. $\frac{5}{6}$ of 132

14. $\frac{3}{4}$ of 244

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Fraction of a Set

OUT OF 18

Warm-Up • 1-12 • 5 | 4 | 5 | 5 | 15 | 5 | 7 | 5 | 6 | 7 | 6 | 5

Core • 13-18 • 12 | 15 | 12 | 20 | 15 | 24

Challenge • 19-22 • 30 | 84 | 51 | 105

5.2 • Of Money and Time

OUT OF 17

Warm-Up • 1-9 • 10 | 10 | 30 | 15 | 10 | 10 | 5 | 5 | 5

Core • 10-17 • 45 | 40 | 27 | 50 | 30 | 30 | 45 | 20

Challenge • 18-19 • 36 | 35

5.3 • Working Backwards

OUT OF 22

Warm-Up • 1-12 • 12 | 12 | 20 | 15 | 30 | 12 | 50 | 25 | 10 | 20 | 75 | 30

Core • 13-22 • 18 | 20 | 25 | 24 | 24 | 80 | 45 | 100 | 100 | 200

Challenge • 23-26 • 63 | 50 | 140 | 60

5.4 • Mixed Fraction Problems

OUT OF 18

Warm-Up • 1-10 • 8 | $\frac{1}{2}$ | $\frac{1}{3}$ | 5 | 4 | 3 | 4 | 11 | 4 | 4

Core • 11-18 • 18 | 20 | $\frac{2}{3}$ | 21 | 22 | 36 | 30 | 18

Challenge • 19-26 • $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{5}{6}$ | $\frac{3}{8}$ | $\frac{17}{12}$ | 35 | 225 | 60

Fri • Fold the Universe, Built

OUT OF 8

Warm-Up • 1-4 • 1 | 2 | 6 | 3

Core • 5-8 • 3 | 16 | 16 | 24

Challenge • 9-14 • 64 | 5 | 15 | 84 | 110 | 183

The Equivalence Proof

Show the same amount two different ways and say why they match.

Warm-Up 8 ITEMS • RECALL

1 $1/2 = ?/4$ — the numerator

2 $2/4$ simplified — type as a/b

3 $1/2 \times 3/3$ — type as a/b

4 $3/6$ simplified

5 $3/3$ as a whole number

6 Multiplying by 1 changes a number by how much

7 $1/2$ of 16

8 $3/4$ of 8

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. $2/3 \times 4/4$ — type as a/b

.....

10. $8/12$ simplified

.....

11. $5/6 \times 3/3$

.....

12. $15/18$ simplified

.....

13. What $3/4$ was multiplied by to become $12/16$

.....

14. $2/3$ of 27

.....

15. $7/10 - 1/5$

.....

Number Line Defence

Every fraction has one place. Put it there and argue for it.

Warm-Up 4 ITEMS • RECALL

1 $\frac{1}{3}$ of 12
_____2 $\frac{1}{2}$ of 18
_____3 $\frac{1}{5}$ of 30
_____4 $\frac{1}{6}$ of 42

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

5. $\frac{3}{8}$ of 88
.....6. $\frac{7}{12} - \frac{1}{4}$
.....

★

Challenge

NOT GRADED · REASON IT OUT

7. 1/2 as a decimal
-
8. 1/4 as a decimal
-
9. 3/4 as a decimal
-
10. 1/5 as a decimal
-
11. 2/5 as a decimal
-
12. 1/10 as a decimal
-
13. 3/8 as a decimal
-
14. 5/8 as a decimal
-
15. 7/10 as a decimal
-
16. Which sits further right: 2/3 or 0.6 — type it
-
17. 3/5 as a decimal
-
18. 7/8 as a decimal
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Mission Review

Everything from six weeks, mixed together.

Warm-Up 10 ITEMS • RECALL

1 $\frac{6}{8}$ simplified — type as a/b

2 $\frac{1}{2}$ of 14

3 $1 - \frac{1}{3}$ — as a/b

4 Larger: $\frac{1}{2}$ or $\frac{1}{3}$ — type it

5 Half of a half — type as a/b

6 Half of a quarter

7 Two quarters

8 Three thirds

9 $\frac{1}{2}$ of 40

10 $\frac{1}{4}$ of 40

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $\frac{3}{4}$ of 32

.....

12. Common denominator of $\frac{5}{6}$ and $\frac{3}{8}$

.....

13. $\frac{2}{5}$ of a number is 10. The number

.....

14. $\frac{1}{2}$ of $\frac{1}{3}$ — type as a/b

.....

15. $\frac{1}{3}$ of $\frac{1}{4}$

.....

16. $\frac{2}{3}$ of $\frac{1}{2}$

.....

17. $\frac{3}{4}$ of $\frac{1}{2}$

.....

18. $\frac{1}{5}$ of $\frac{1}{2}$

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS • RECALL

1 $\frac{2}{4}$ simplified — type as a/b

2 $\frac{1}{4}$ of 16

3 Is $\frac{3}{6}$ the same as $\frac{1}{2}$ — yes or no

4 $\frac{2}{3}$ of 9

5 $\frac{1}{4}$ of 32

6 $\frac{1}{3}$ of 24

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

7. $\frac{12}{16}$ simplified

.....

8. $\frac{5}{6}$ of 24

.....

9. Larger: $\frac{2}{3}$ or $\frac{3}{4}$ — type it

.....

10. $\frac{5}{8}$ of 64

.....

11. $\frac{2}{3}$ of 42

.....

★

Challenge

NOT GRADED · REASON IT OUT

12. $\frac{1}{3} + \frac{1}{3}$ — as a/b
-
13. $\frac{1}{5}$ as a decimal
-
14. $\frac{5}{6} - \frac{1}{6}$ — as a/b
-
15. $\frac{1}{2} + \frac{1}{3}$ — as a/b
-
16. $\frac{3}{8}$ as a decimal
-
17. $\frac{2}{3} + \frac{3}{4}$ — as a/b
-
18. After spending $\frac{3}{5}$, \$20 is left. The starting amount
-
19. $\frac{1}{2} + \frac{1}{4}$
-
20. $\frac{5}{6} - \frac{1}{6}$
-
21. $\frac{3}{8} + \frac{1}{8}$
-
22. $\frac{3}{4} + \frac{1}{6}$
-
23. $\frac{7}{8} - \frac{1}{2}$
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Mission 04 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 2 ITEMS • RECALL

1 $\frac{4}{8}$ simplified — type as a/b

2 $\frac{1}{2}$ of 20

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

3. $\frac{1}{2} = \frac{?}{10}$ — the numerator

4. $\frac{10}{15}$ simplified

5. Larger: $\frac{3}{5}$ or $\frac{5}{8}$ — type it

6. $\frac{2}{3}$ of 18

7. Common denominator of $\frac{2}{3}$ and $\frac{3}{4}$

8. $\frac{3}{4}$ of 44

 **Challenge** NOT GRADED • REASON IT OUT

9. $\frac{1}{4} + \frac{1}{4} = \frac{a}{b}$

10. $\frac{1}{2} + \frac{1}{4}$ — as a/b

11. $\frac{3}{4} - \frac{1}{3}$ — as a/b

12. $\frac{3}{4}$ as a decimal

13. $\frac{2}{3} + \frac{3}{4}$ — as $\frac{a}{b}$

14. $\frac{3}{8}$ of a number is 9. The number

15. $\frac{1}{4} + \frac{1}{4}$

16. $\frac{2}{3} + \frac{1}{4}$

17. $\frac{5}{6} + \frac{7}{12}$

18. $\frac{7}{8}$ of 176

19. $\frac{2}{3}$ of $\frac{3}{5}$ — type as a/b

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 6 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

6.1 • The Equivalence Proof

OUT OF 15

Warm-Up • 1-8 • 2 | $\frac{1}{2}$ | $\frac{3}{6}$ | $\frac{1}{2}$ | 1 | 0 | 8 | 6

Core • 9-15 • $\frac{8}{12}$ | $\frac{2}{3}$ | $\frac{15}{18}$ | $\frac{5}{6}$ | 4 | 18 | $\frac{1}{2}$

Challenge • 16-26 • $\frac{2}{3}$ | $\frac{2}{3}$ | 1 | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{5}{6}$ | $\frac{5}{12}$ | $\frac{7}{6}$ | $\frac{17}{12}$ | 95

6.2 • Number Line Defence

OUT OF 6

Warm-Up • 1-4 • 4 | 9 | 6 | 7

Core • 5-6 • 33 | $\frac{1}{3}$

Challenge • 7-18 • 0.5 | 0.25 | 0.75 | 0.2 | 0.4 | 0.1 | 0.375 | 0.625 | 0.7 | $\frac{2}{3}$ |
0.6 | 0.875

6.3 • Mission Review

OUT OF 18

Warm-Up • 1-10 • $\frac{3}{4}$ | 7 | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{2}$ | 1 | 20 | 10

Core • 11-18 • 24 | 24 | 25 | $\frac{1}{6}$ | $\frac{1}{12}$ | $\frac{1}{3}$ | $\frac{3}{8}$ | $\frac{1}{10}$

Challenge • 19-26 • $\frac{1}{2}$ | 0.5 | $\frac{11}{12}$ | $\frac{5}{12}$ | 1 | 35 | $\frac{1}{2}$ | $\frac{1}{2}$

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{2}{3}$ | 4 | yes | 0.2 | $\frac{2}{3}$ | $\frac{3}{4}$ | 6 | $\frac{2}{3}$ | 8 | $\frac{1}{2}$ | 8

Core • 13-22 • $\frac{5}{6}$ | $\frac{3}{4}$ | 20 | $\frac{3}{4}$ | 0.375 | $\frac{11}{12}$ | $\frac{3}{8}$ | 40 | $\frac{7}{10}$ | 28

Challenge • 23-26 • $\frac{17}{12}$ | 50 | $\frac{29}{24}$ | 84

Fri • Mission 04 Test

OUT OF 14

Warm-Up • 1-4 • $\frac{1}{2}$ | $\frac{1}{2}$ | 10 | $\frac{1}{2}$

Core • 5-14 • 5 | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{5}{12}$ | $\frac{5}{8}$ | 12 | 0.75 | 12 | $\frac{11}{12}$ | 33

Challenge • 15-19 • $\frac{17}{12}$ | 24 | $\frac{17}{12}$ | 154 | $\frac{2}{5}$